

EXPERIMENT 4

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QUESTION A:

You are given a table `tbl_happiness` that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order.

QUERY:

```
SELECT sno, rankings, country,
CASE
WHEN COUNTRY = 'India' THEN 1
WHEN COUNTRY = 'Sri Lanka' THEN 2
ELSE 3
END AS DERIVED
FROM tbl_happiness
ORDER BY DERIVED, RANKINGS;
```

RESULT:

	sno [PK] integer	rankings integer	country character varying (50)	derived integer
1	8	126	India	1
2	9	128	Sri Lanka	2
3	1	1	Finland	3
4	2	2	Denmark	3
5	3	3	Iceland	3
6	4	4	Israel	3
7	5	5	Netherlands	3
8	6	6	Sweden	3
9	7	7	Norway	3

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QUESTION B:

Make all the rankings in descending order except India and Sri Lanka.

QUERY

```
SELECT *  
FROM tbl_happiness  
ORDER BY  
    CASE  
        WHEN Country = 'India' THEN 1  
        WHEN Country = 'Sri Lanka' THEN 2  
        ELSE 3  
    END,  
    CASE  
        WHEN Country IN ('India', 'Sri Lanka') THEN Rankings  
        ELSE -Rankings  
    END;
```

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RESULT

	sno [PK] integer	rankings integer	country character varying (50)
1	8	126	India
2	9	128	Sri Lanka
3	7	7	Norway
4	6	6	Sweden
5	5	5	Netherlands
6	4	4	Israel
7	3	3	Iceland
8	2	2	Denmark
9	1	1	Finland

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QUESTION C:

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

QUERY:

```
INSERT INTO Employees (employee_id, name, reports_to, age)
VALUES
(9, 'Hercy', NULL, 43),
(6, 'Alice', 9, 41),
(4, 'Bob', 9, 36),
(2, 'Winston', NULL, 37);

SELECT e1.employee_id, e1.name, count(e2.reports_to) as report_count, round (AVG(e2.age))
FROM Employees as e1
join Employees as e2
on e1.employee_id = e2.reports_to
group by e1. employee_id;
```

out Messages Notifications

RESULT:

	employee_id [PK] integer	name character varying (50)	report_count bigint	round numeric
1	9	Hercy	2	39